The Science, and How to Show It to the Judges

Team Shay Rafe3 · NASA Space Apps 2026 · "Space Mission Design Game" · Pitch Sun 15 Nov, 16:15 (submit by 15:15) · for the whole team

> **Our message in one sentence:**
> "Every choice in a space mission pulls on every other one. Our game makes you feel that with NASA's real numbers, and when your mission fails, it tells you the physics of why."

The shift: judges don't score the game. They score the *problem, the science, the NASA data and how clearly we show it*. The game is our **proof**: a live demo of the science. Every slide, every demo click and every answer should lead back to one of the science ideas in section 3.

1. How the judges score (5 criteria, 1–5 each, equal weight)

Criterion	Judge's question	What gets a 5	How the **science** wins it for us
Impact	How big is the problem? Who does it help?	Major problem, wide reach	Students learn orbits, power and radio separately. We teach the **trade-offs between them**, free in a browser, in plain words
Creativity	New idea or real improvement?	A fresh approach	Science you *feel*: the rocket equation punishes extra mass; sunlight dies at Saturn. Team roles negotiate real constraints
Validity	Scientifically sound? Feasible?	Fully developed, correct	Real formulas, every number sourced, invented numbers labelled. Our flight times match Kepler's law (section 3.7)
Relevance	Answers the challenge? NASA data at the core?	NASA data central, not decoration	Remove NASA data and the physics breaks: no real gravity, sunlight, storms or margins. Name each dataset exactly (section 4)
Presentation	Clear, engaging, structured?	Judges get it at once	One idea per slide, a number on screen, a live "fail → why → fix" moment

> **Rules that disqualify:** max **7 slides** (title included) **or** a video of max **30 s** · public links, no login · English (+ subtitles on video) · **AI disclosure** · nobody under 18 in the video · name NASA data exactly (at least one, or no Global Award) · list every outside resource.

2. The science in one picture: every world teaches one constraint

World	The one idea	The number that proves it	Real mission
Moon	**Fuel vs mass** (rocket equation)	1.7 kg of fuel for every 1 kg of ship to land	Apollo 11, CAPSTONE
Mars	**Air is free braking**; night; radio delay	Heat shield + parachute: 0.12 kg fuel per kg instead of 1.8	Perseverance
Venus	**Pressure and heat** limit lifetime	92 bar, 464 °C: the lander lives 127 minutes	Venera 13
Mercury	**Too much sunlight**; hard to stop	9,083 W/m² (6.7× Earth); 2.75 kg fuel per kg	MESSENGER, BepiColombo
Jupiter	**Radiation**	Juno's 172 kg vault cuts radiation 800×	Juno

Saturn	**Power and data far away**	1 kW solar array gives ~8 W; one nuclear MMRTG = 14 arrays	Cassini
Uranus, Neptune	**Time and the radio link**	12–16 years; a big dish sends ~3 Gb a month from Neptune	Voyager 2

3. The science cards (learn these, every speaker)

Each card: the idea · the formula · real numbers and source · a worked example with our game's numbers · the moment to show it · the judge question we expect.

3.1 The rocket equation: why mass is everything VALIDITY CREATIVITY

Idea	A rocket pushes by throwing fuel out the back. Fuel also has to push *itself*, so the fuel needed grows **exponentially** with the speed change (Δv).
Formula	`Δv = Isp · g₀ · ln(m_full / m_empty)` → `fuel = m_dry · (e^(Δv / (Isp·g₀)) − 1)` · Isp 300 s, g_0 9.81 m/s² → exhaust speed 2.94 km/s
Source	NASA GMAT default thruster Isp = 300 s · NASA Glenn: rocket thrust
Example	Moon landing needs Δv ≈ 2.93 km/s → **1.71 kg fuel per 1 kg of ship**. Add 100 kg of instruments → you also need 171 kg more fuel → a bigger rocket.
Show it	In the builder, add one instrument and watch the FUEL and BOOSTER gauges both move. Launch too heavy → "TOO HEAVY", explained.
Judge asks	**"Why not just add more fuel?" → Fuel is mass too. Past a point each extra kg of fuel mostly lifts other fuel. That's the exponential.**

3.2 The inverse-square law: why solar dies far from the Sun VALIDITY

Idea	Sunlight spreads over a sphere. Twice as far → a quarter of the light. Arrays also wear out.
Formula	`P = P_1AU / d² · 0.95^years · usable fraction` (d in AU)
Source	Sunlight from NASA Planetary Fact Sheets (Earth 1,361, Mars 586, Saturn 14.8 W/m²) · 5%/yr ageing = GMAT default · arrays ≈30 W/kg NASA SST 2026 ch. 3
Example	A 1 kW array (33 kg) gives **335 W at Mars, 29 W at Jupiter, 8 W at Saturn, 0.6 W at Neptune**.
Show it	Same ship, change destination Mars → Saturn: the POWER lamp goes red. Fly it anyway → "OUT OF POWER... sunlight falls with distance squared."
Judge asks	**"Juno uses solar at Jupiter. Is your game wrong?" → No: Juno needed three 9 m wings for ~400 W. Saturn gets over 3× less light than Jupiter, so Cassini used nuclear power.**

3.3 Nuclear power (MMRTG) vs solar RELEVANCE

Idea	Radioactive decay makes heat → electricity, day and night, at any distance. Heavy, expensive, slow to approve.
Numbers	110 W, 45 kg, 14-year design life · NASA MMRTG fact sheet
Example	At Saturn after Cassini's 6.7-year trip: **one 45 kg MMRTG = 14 solar arrays (~470 kg)**. That is why Cassini, Perseverance and Curiosity are nuclear.
Judge asks	**"Why does nuclear add development time?" → Real launches of nuclear power need a safety approval that takes years; the game adds 300 days (labelled game rule).**

3.4 Batteries and night VALIDITY

Formula	`energy needed (Wh) = (power used − nuclear power) × dark hours`, then +35% margin
Numbers	Mars night 12.3 h · Venus lander 127 min on batteries · Li-ion 150–270 Wh/kg (NASA SST 2026)
Example	A 200 W rover through a Mars night: 2,460 Wh → with margin 3,321 Wh → two 2 kWh batteries.

3.5 Radio link and light delay: why the data may never come home VALIDITY RELEVANCE

Idea	The radio signal also spreads with distance², so the data rate falls with distance². Light takes 8.3 min per AU, so far missions must run themselves.
Formula	`rate = dish rate at 1 AU / d²` · `data sent = rate × contact time` (8 h a day, game rule) · `delay = distance / speed of light`
Source	MRO 3 m dish 6 Mbps from Mars (NSSDCA MRO) · LRO Ka-band 100–300 Mbps (NSSDCA LRO) · delay (NASA Goddard)
Example	Big dish, 30 days: **Mars 1,152 Gb · Saturn 29 Gb · Neptune 2.8 Gb**. Delay: Mars 4–24 min, Saturn ~80 min, Neptune ~4 h. Real: Voyager 2 sent 21.6 kbps from Neptune.
Show it	A cheap antenna at Saturn → the report says how much science was lost. The "light delay" line is why landers land themselves.

3.6 NASA's design margins (JPL Concept Maturity Levels) RELEVANCE VALIDITY

Idea	Early designs always grow. JPL demands spare capacity before a concept is accepted: margin = safety against the unknown.
Numbers	CML 4: **35%** mass, power and energy margin · telecom **3 dB** above minimum for deep space (6 dB near Earth) · NASA Discovery 2019: **25%** cost reserve · JPL CML paper
Explain dB	3 dB = $10 \cdot \log_{10}(2)$ = **twice** the signal power. In the game: room to send 2× the data.
Show it	A gauge is green only with the margin kept. UNDER = risky, WELL USED = margin kept, OVER = wasted mass and money. This is how we reward **efficient resource use** (named in the challenge).
Judge asks	**"Why 35%?" → It is JPL's number for this design stage, from the paper in the challenge's own resources.**

3.7 Orbits and flight paths (Hohmann transfer, gravity assists) VALIDITY

Idea	The cheapest path between two orbits is half an ellipse touching both (Hohmann). Flying past planets (gravity assist) saves fuel but takes longer.
Formula	Kepler's 3rd law: `T = a^1.5 years` (a in AU) → transfer time = half of that, with $a = (r_1 + r_2)/2$
Check	Our maths: Mars 259 days, Jupiter 2.7 yr, **Saturn 6.05 yr**. Real: Perseverance 203 days (faster path, more energy), Cassini 6.7 yr with 4 flybys. · NASA Basics of Space Flight ch. 4
Show it	Pick the "fast" path: shorter trip but the rocket lifts less. Trade time against mass.

3.8 Landing: gravity and air VALIDITY

Numbers	Gravity: Moon 1.62, Mars 3.73, Venus 8.87 m/s² (NASA fact sheets). Perseverance's parachute: 1,512 → 306 km/h = **420 → 85 m/s**, the same numbers the game uses.
Idea	No air (Moon, Mercury): only the engine can stop you. Thin air (Mars): heat shield + parachute do most of it. Thick air (Venus): the air does it all, the pressure is the enemy.
Example	Mars with heat shield + parachute: **0.12 kg fuel per kg**. Without using the air: **1.8 kg per kg**, 15× more.
Show it	Hand-fly the Mars landing: real gravity, a brake light, a safe-speed gauge.

3.9 Space weather: real solar storms (NASA DONKI) RELEVANCE

Idea	The Sun throws clouds of charged particles. The Kp index (0–9) measures the geomagnetic storm; Kp 9 is the top of NOAA's G5 "extreme".
Data	15 real geomagnetic-storm records from 2024 (DONKI GST), incl. **Kp 9 on 10 May 2024**, the strongest storm in ~20 years · api.nasa.gov (DONKI) · NOAA scales · NASA on the May 2024 storm
Logic	A real storm is drawn. Kp ≥ 7.5 → 70% chance to damage an instrument or antenna unless shielded (15% for weaker storms; chances = game rule).
Show it	The cruise event card prints the real date and Kp. Real precedent: a 2003 storm killed the MARIE instrument on Mars Odyssey.

3.10 Heat, pressure and radiation CREATIVITY

Numbers	Mercury sunlight 9,083 W/m² → MESSENGER's ceramic sunshade · Venus 92 bar, 464 °C → Venera 13 lived 127 min · Jupiter's belts → Juno's 172 kg titanium vault
Logic	Over 5,000 W/m² with no sunshade → "OVERHEATED" · no pressure hull on Venus → crushed · no shielding at Jupiter → parts fail (thresholds = game rule, anchored on the missions)

3.11 The mission process (how NASA designs) RELEVANCE IMPACT

Idea	"Begin with the end in mind": goal → requirements → design → build → operate. 4 segments: Launch, Space, Ground, User. A design team of specialists must agree.
Source	NASA Design and Develop Science Missions · Goddard Mission Design Lab
In game	Goal first (pick world and task) → 4 roles: Planner ≈ Launch, Engineer ≈ Space, Comms ≈ Ground, Scientist ≈ User → GO / NO-GO review like a design review.

4. NASA data, named exactly (say these words on slide 4)

Say exactly	What it does	What breaks without it
NASA **DONKI** geomagnetic storm (GST) records, 2024, via api.nasa.gov	Real storms hit the ship during cruise	No real space weather
NASA **NSSDCA Planetary Fact Sheets**	Gravity, sunlight, distance for 8 worlds	Landing, power and radio physics become fake
JPL **Concept Maturity Levels** (Wessen et al., 2013)	35% margins, 3 dB link, GO / NO-GO	No standard for "good design"
NASA **GMAT** default values	Isp 300 s, 5%/yr solar ageing	Fuel and power numbers unanchored
NASA **MMRTG** fact sheet	Nuclear power part	No outer-planet missions
NASA **State of the Art of Small Spacecraft Technology** (2026)	Solar W/kg, battery Wh/kg	Part masses invented
NASA NSSDCA: **MRO, LRO**; MRO **CTX, CRISM**; Cassini **MAG**	Dishes, data rates, instruments	Science and data invented
Mission facts: Apollo 11, CAPSTONE, Perseverance, Cassini, Juno, MESSENGER, Voyager 2 (NASA)	Flight paths, times, braking burns, parachute	No real-mission comparison
Partner agency: **ESA** Venus Express, BepiColombo	Venus and Mercury paths	List under "Space Agency Partner data"

Full list with links: **NASA_Data_In_Game.pdf** (on the website). Rocket prices/lift come from makers and Wikipedia: list them under "other references".

5. The 7 slides: science first (one story, also the 30 s video)

#	Slide	Scores	What to put on it	Time
1	Title	Presentation	Name, team, exact challenge title, Ramallah · the one-sentence message	0:15
2	The problem	Impact	One student, one sentence: learned orbits and rockets separately, never a whole mission. **+ one sourced number (team to find; don't invent)**	0:30
3	The science idea	Creativity	"Everything pulls on everything": the trade-off web (mass ↔ fuel ↔ rocket ↔ budget; distance ↔ power ↔ data). 1 picture	0:25
4	How it works + NASA data	Relevance, Validity	Data → formula → consequence. Exact dataset names (section 4). Two formulas: rocket equation and $1/d^2$. One number: "1 kW of solar = 8 W at Saturn"	0:35

5	Demo	Validity, Presentation	Live or GIF: build → fly to Saturn on solar → "OUT OF POWER, sunlight falls with distance^2" → swap in MMRTG → success. QR to the website	0:40
6	Impact	Impact	Who: students, teachers, space clubs; free, browser, no login; 4 players learn teamwork like NASA's design labs. **+ a playtest quote (do one!)**	0:20
7	Limits & next	Validity	Honest limits (section 7) · next: real GMAT runs, classroom pilot, Arabic version · team names	0:15

> **30 s video** = slides 1 → 2 → 4 → 5, ~7 s each, English subtitles, game footage only. **Live pitch:** 3:00 + 2:00 Q&A · one speaker per part · one person drives the demo · offline recording as backup · rehearse 3× with a timer.

6. Judges' questions: answer in 3 beats (answer → evidence → impact), max 20 s

Question	Our answer
Is the physics real or made up?	Real formulas (rocket equation, inverse-square, Kepler). Every number sits in one file with its source; invented ones are labelled "game rule". Our Saturn transfer time matches Kepler's law: 6.05 yr.
Which NASA data, and what breaks without it?	Section 4: DONKI, Planetary Fact Sheets, JPL CMLs, GMAT, MMRTG, SST 2026, NSSDCA. Without them gravity, sunlight, storms and margins are fake.
Did you use GMAT?	We use GMAT's default values (Isp, solar ageing). Times and Δv come from real missions and Kepler maths. Running GMAT itself is our next step. *(Never say "computed with GMAT".)*
What does a student learn?	That a mission is a system: one extra instrument changes fuel, rocket, power, data and cost. And the physics behind each failure.
How is this different from Kerbal Space Program?	KSP is flight. We're mission design: budget, power, data, comms, time, NASA margins, a team of 4 roles, and an explanation of why it failed.
Why 4 roles?	NASA designs missions in teams of specialists (Goddard's Mission Design Lab). Each role owns part of the ship and must negotiate and sign off.
Who will use it?	Students and teachers anywhere, free in the browser. Plain words, three difficulty levels per world.
What's the hardest science in it?	The coupling: the rocket equation makes every kilogram cost fuel, and the inverse-square law makes distance cost power and data, both at once.

7. Real vs game rule (be honest: it scores Validity)

Real (sourced)

- Gravity, sunlight, distances, light delay
- Rocket equation, Isp 300 s
- Part specs: MMRTG, solar, batteries, MRO, LRO, CTX, CRISM, MAG
- JPL margins, 25% cost reserve
- Flight times and braking burns of real missions
- 2024 DONKI storms; 20 event stories from real missions

Game rule (labelled)

- Part prices, budgets, time limits
- 8 h/day of radio contact
- Landing-leg 4 m/s, airbag 12 m/s limits
- Damage chances in events
- Development days; scores and grades
- Robotic missions only; no full GMAT runs

8. Before 15:15 Sunday: checklist

- ☐ Everyone can say the one-sentence message and explain cards 3.1, 3.2, 3.5, 3.6 without notes
- ☐ **One sourced number for the problem slide** and **one playtest quote** for the impact slide
- ☐ 7 slides max, built from section 5 · 30 s video with English subtitles · public links
- ☐ Demo path rehearsed: Saturn on solar fails → MMRTG fixes it · offline recording as backup
- ☐ GitHub Pages turned on; website + game link open with no login · team names on the website

- ☐ Submission: NASA data list (section 4), partner data (ESA), other references, AI disclosure